Adya Thikkavarapu S

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.55
FIITJEE Junior College	Class XII (TSBIE)	2024	96.60%
Manthan School	Class X (IGCSE)	2022	95.10%

SCHOLASTIC ACHIEVEMENTS

- **100%** in Mathematics, Physics and Chemistry in Class XI TSBIE Board
- International Certificate of Education award result of **Distinction** in the IGCSE 2022 exam series
- Scored A* in all subjects
- **Gold Scholar** in the Ei Asset Talent Search Test 2018
- Ranked Top 50 in India

RELEVANT COURSES

Computer Science: Introduction to Programming, Introduction to Computing, Discrete Maths for Computer Science, Operating Systems 1, Data Structures and Applications, Algorithms, Theory of Computation, Compilers 1, Artificial Intelligence

Mathematics: Calculus I and II, Probability and Random Variables, Elementary Linear Algebra, Introduction to Statistics, Differential Equations

SKILLS

Programming Languages: C/C++, Python

Frameworks/Libraries: Pytorch, Torchvision, Hugging Face (Transformers), Scikit-Learn, Matplotlib, NumPy

CS Fundamentals and Tools: DSA, OOP, Git

PROJECTS

Deepfake Detection Framework: Comparing CNN and ViT Architectures

- Built a deep learning framework in PyTorch comparing CNN and Vision Transformer architectures for facial deepfake recognition.
- **XAI Audit:** Audited model decision making by utilizing **Grad-CAM** for the CNN and **Attention Rollout** matrices for the ViT to visually identify regions of interest.
- Quantitatively evaluated the models by plotting confusion matrices and calculating accuracy.
- Methodologies used: Explainable AI (XAI), Data Augmentation, Gradient Tracking (Hooks), Model Auditing.

QLoRA Implementation From Scratch

- Built a custom **QLoRA (Quantized Low Rank Adaptation)** training pipeline for LLMs using PyTorch from scratch, implementing the underlying matrix computations.
- Reduced the number of trainable parameters of the model by over 97% in the target layers by injecting custom LoRA layers.
- Implemented 4-bit NormalFloat (NF4) post-training quantization.
- Fine-tuned 125M transformer model on the task of sentiment analysis and successfully reduced validation perplexity

Reinforcement Learning Agent | Q-Learning Implementation

- Developed and trained a Q-learning agent from scratch on a discrete FrozenLake-v1 Gymnasium environment.
- Optimized the agent's action-value function by implementing an epsilon-greedy exploration-exploitation strategy and a temporal-difference update policy.

- Visualized and evaluated agent performance by plotting reward convergence over 1,000 training episodes.

Implementing Attention Layers & Sentiment Analysis Models:

- Implemented Bahdanau, Luong Dot, Luong General and Luong Concat attention mechanisms as stand-alone layers compatible with PyTorch RNN and LSTM outputs.
- Integrated attention layers with 4 base recurrent networks (Vanilla RNN, LSTM, BiRNN, BiLSTM), evaluating the 12 models in the task of sentiment analysis.
- Built a custom sequence-processing pipeline, including tokenization, text padding/truncation and integration of pretrained GloVe word embeddings.

POSITIONS OF RESPONSIBILITY

Coordinator at Elan & nVision, IIT Hyderabad

Core at Shuffle, Dance Club of IIT Hyderabad

EXTRACURRICULAR

- Represented IITH in Inter IIT Cultural Meet 8.0 and other inter-collegiate competitions as part of the Dance team.
- Conducted dance workshops within the IITH community as a part of the dance club.
- Management: worked in the *Productions and Pronites* domain of IITH's techno-cultural fest, Elan & nVision